App Stores as Revenue Oracles

Architecture white paper | AppLov | Revised November 2026

Executive summary

AppLov proposes using official Apple App Store and Google Play reporting as authoritative evidence of app performance and platform-attributed revenue. An off-chain verification service reconciles store reports, payout records, refunds, taxes, and fees, then publishes signed accounting results for on-chain revenue-sharing contracts. Investors hold tokens or NFTs that encode contractual economic rights, subject to applicable law.

Why app stores matter

App stores provide an independent reporting channel relative to a developer's own dashboard. Their APIs and financial reports can support verification of downloads, releases, subscriptions, proceeds, and adjustments, but they are not immutable, uniformly real-time, or a complete view of all app revenue. Data access, reporting granularity, permissions, and settlement timing vary by platform.

Proposed architecture

A Flutter-based web portal communicates with Firebase Data Connect for identity-linked application records, funding rounds, and reconciled financial statements. A backend ingestion/oracle service authenticates to App Store Connect and Google Play reporting interfaces, normalizes reports, and posts signed attestations to smart contracts. Smart contracts cannot directly call ordinary HTTPS store APIs. A separate payment rail handles fiat settlement and optional conversion to USDC before contract funding.

Revenue distribution

Each eligible app has a distinct accounting ledger and distribution pool. Store financial reports are matched to deposits, fees, refunds, chargebacks, and currency conversion. After funds are actually received and a reconciliation period closes, an authorized service funds the app's distribution contract. Token-holder claims are calculated according to the legal revenue-sharing agreement and a defined snapshot or accrual policy.

Financial routing

A financial routing provider may be used if its capabilities and the app-store payout configuration permit the intended arrangement. Neither Apple nor Google should be assumed to support arbitrary per-app split payouts to a smart contract. A practical fallback is receipt into a controlled bank account, per-app reconciliation, and subsequent transfer into a compliant stablecoin payment flow.

Trust and controls

Use least-privilege API access, encrypted credentials, signed attestations, audit logs, app-specific ledger entries, independent reconciliation, dispute windows, emergency pauses, and transparent corrections. Clearly distinguish provisional sales estimates from finalized proceeds and funds actually received.

Legal considerations

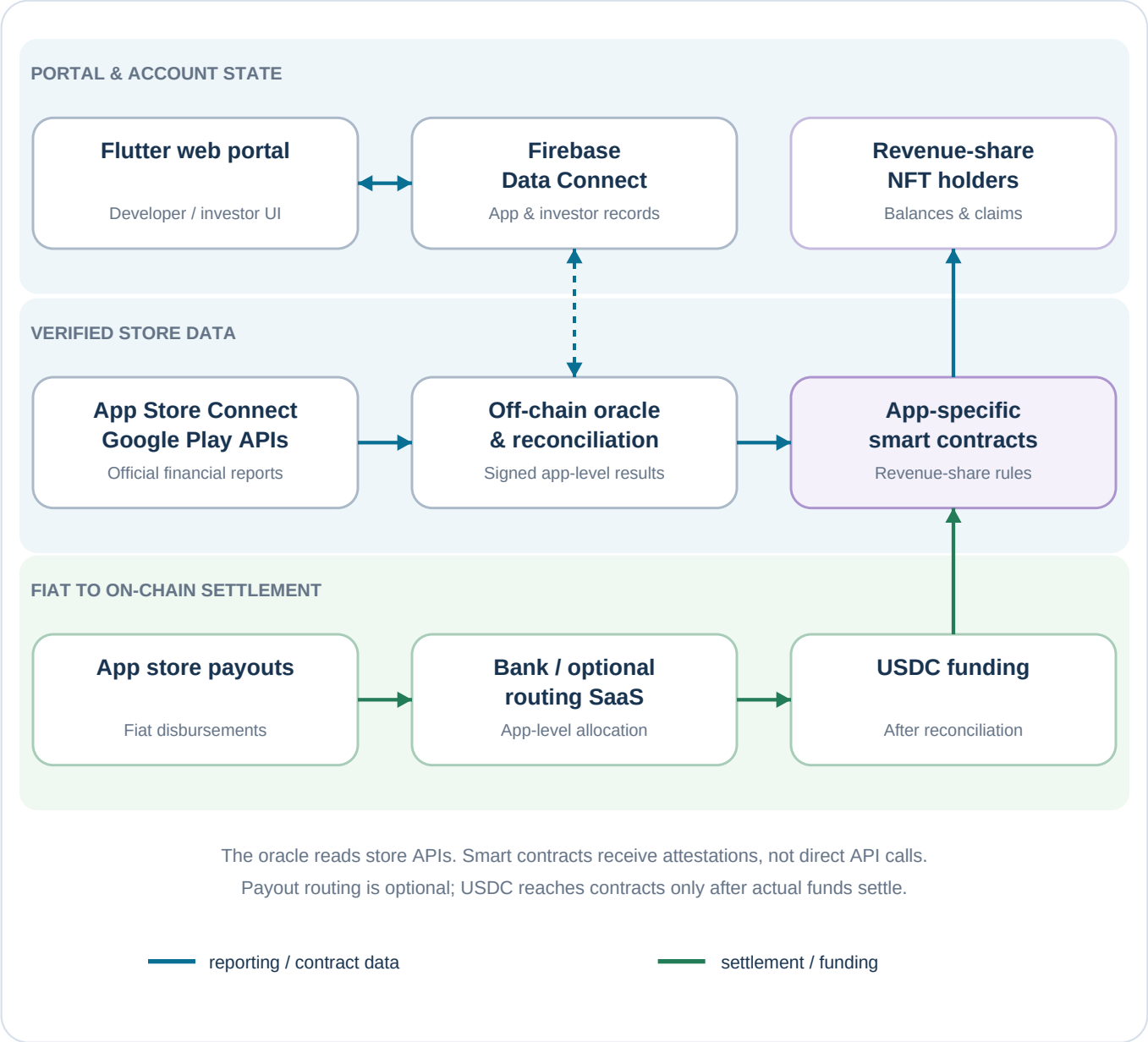
Revenue-share NFTs may constitute securities, regardless of their token format. Offering restrictions, investor eligibility, disclosures, custody, money transmission, tax treatment, and secondary trading require legal review. Token ownership is not automatically equity ownership.

Implementation roadmap

Phase 1: read-only store reporting integration and per-app dashboard. Phase 2: reconciled revenue ledger and testnet distribution using simulated USDC. Phase 3: production funding and payout pilot with compliance controls. Phase 4: evaluate routing providers and restricted secondary transfers.

System architecture flowchart

Data verification and movement of funds are separate paths that converge at the app-specific smart contract.



The oracle reads store APIs. Smart contracts receive attestations, not direct API calls.
Payout routing is optional; USDC reaches contracts only after actual funds settle.